



PROMISEDLAND.COM

Linux Deployment Proposal

To: Stakeholders, PromisedLand.com

Subject

Linux Server Deployment Plan
— Independence, Missouri Data Center

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Executive Summary

This proposal outlines a robust, scalable Linux infrastructure designed to support PromisedLand.com's new data center in Independence, Missouri. As the organization expands its hybrid call center operations across North America, the implementation of a centralized Ubuntu Server on AWS EC2 will provide a high-availability environment for 38 local employees.

The primary objective of this deployment is to modernize IT operations by transitioning to a non-proprietary, secure file-sharing architecture that mirrors the company's organizational structure. Key features of this plan include:

- **Strategic System Design:** A centralized file server role configured to reduce data silos and improve cross-departmental redundancy.
- **Advanced Access Control:** A group-based permission model integrated with Access Control Lists (ACLs) to ensure data integrity while meeting specific stakeholder needs, such as read-only access for the CIO.
- **Security & Stability:** A multi-layered security approach featuring a hardened firewall (UFW), static IPv4 networking for consistent connectivity, and automated maintenance via cron-scheduled reboots.
- **Efficient Storage Management:** A structured disk partition layout using the ext4 filesystem, optimized for both current performance and future storage expansion.

Adopting this Linux-based framework guarantees that PromisedLand.com will achieve a professional, cost-effective, and highly secure foundation that is ready for immediate deployment and long-term growth.

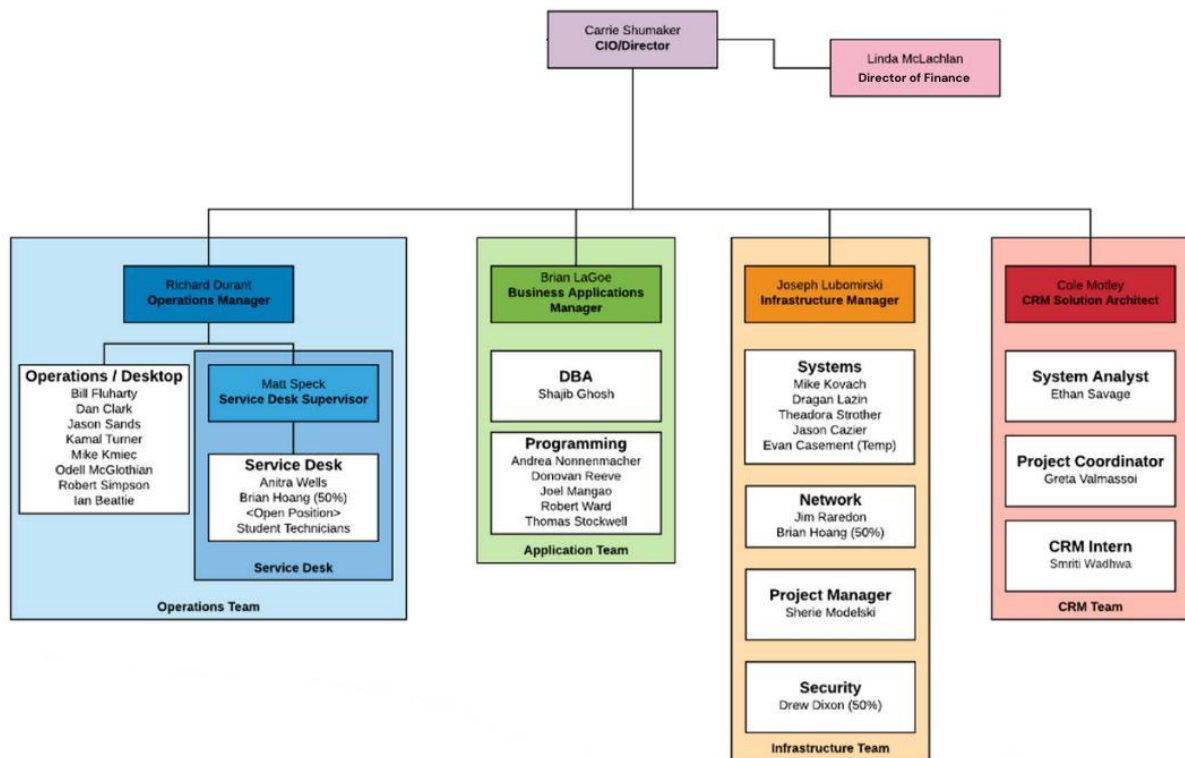


Fig.1 PromisedLand’s Organizational Chart [Source: <https://byupw.instructure.com/courses/40835/files/5974775?wrap=1>]

Server Roles & System Design

To support PromisedLand’s expansion into the Independence, Missouri data center, I am proposing the deployment of a Linux-based File Server utilizing Ubuntu Server on AWS EC2. This server will serve as the central repository for the facility’s 38 employees, ensuring that the Infrastructure, Operations, Applications, and CRM teams have high-availability access to critical documents and departmental resources.

- **Primary Function:** Centralized file storage and management for all local departments.
- **Infrastructure Integration:** The server provides a scalable, non-proprietary foundation that improves redundancy for hybrid call centers across the U.S. and Canada.

Centralizing file management will reduce data silos and provide a scalable foundation that can be easily replicated as PromisedLand continues its growth across North America. This design prioritizes reliability and stakeholder needs by providing a stable, high-performance environment for hybrid call center operations.

Video Demonstration: In my demonstration, the *hostnamectl* and *systemctl* status commands will be used to verify the server's identity and active file services.

Disk Partition Layout

To ensure system efficiency and prevent a single partition from crashing the operating system, a structured partitioning scheme will be implemented. Additionally, to ensure system stability and allow for future expansion without downtime, the storage architecture will utilize a structured partitioning scheme. The following layout is designed to prevent a single partition (such as user data) from filling up and crashing the entire operating system.

- **Layout:** The layout will use the ext4 filesystem for its proven reliability and performance in Linux environments. The */srv/data* partition is specifically separated to allow for independent scaling as PromisedLand's data needs grow.

Verification: The *lsblk* and *df -h* commands will be demonstrated to prove that partitions are correctly formatted and mounted to the filesystem.

Users and Groups

The system will mirror PromisedLand's organizational chart to maintain strict access control. Users will be organized into primary groups based on their functional departments:

- **Group Structure:** Primary groups such as infrastructure, operations, applications, crm, and executive will be created to manage system access.
- **User Mapping:** Users like Richard Durant and Matt Speck will be assigned to the operations group.
- **Administrative Access:** Sudo privileges will be restricted to key personnel, such as Joseph Lubomirski (Infrastructure Manager), to ensure system integrity.

Each user will have a unique account to ensure accountability and provide a personalized environment for their specific job functions.

Video Demonstration: The *id* and *groups* commands will verify that users are assigned to the correct departmental groups.

Shared File Storage & Permissions

Collaboration will be facilitated through shared directories under */srv/data*, managed by group-based permissions. This ensures that security is maintained.

- **Group Permissions:** Directories will use *chmod* and *chown* to ensure only relevant teams (e.g., the Operations Team) can access support documents.
- **Departmental Folders:** Folders such as */srv/data/operations* will be owned by the operations group with 770 permissions (read/write/execute for the group, no access for others).
- **CIO Access:** As required, Carrie Shumaker will be granted *read-only access* to all group files. This will be implemented using Access Control Lists (ACLs) to provide specific viewing rights without granting write permissions.

Verification: An *ls -l* command will be used in the video to display the ownership and permission strings for these directories.

Networking & Automation Configuration

- **Static IPv4 Configuration:** To ensure the PromisedLand file server remains reachable at a consistent address for all 38 employees, the system will be configured with a Static IPv4 address. This prevents connectivity issues that can occur with dynamic addressing (DHCP) and allows for reliable firewall rule mapping.
- **Networking Verification:** During the video demonstration, the network interface and static assignment will be verified using the *ip a* command to show the active address and *cat /etc/netplan/*.yaml* to display the persistent configuration file.
- **Scheduled Restarts (Automation):** To maintain long-term system health and prevent memory leaks, the server is configured for a scheduled restart every 2 years. This is handled via the system's *cron daemon*, which automates maintenance tasks without manual intervention.

Automation Verification: The configuration will be proven in my demo video by running *crontab -l*, which will display the specific *cron job* string containing the reboot command and timing logic.

Security & System Stability

Security is a critical component of this proposal, utilizing multiple layers of protection. Security is integrated at every level of this deployment to protect PromisedLand's digital assets:

- **Networking:** The server uses a **Static IPv4 configuration** to ensure consistent access for all departments and to provide a stable target for security policies. Verification will be shown via *ip a* and *cat /etc/netplan/*.yaml*.
- **Firewall:** The Uncomplicated Firewall (UFW) will be installed and configured to protect the system by allowing only authorized traffic (SSH on port 22 and departmental file protocols). Status will be verified using *ufw status*.
- **System Updates:** To keep the system secure, a non-root user will be used for daily tasks, and *apt update* && *apt upgrade* will be run regularly to ensure all packages are current, ensuring that all software is patched against known vulnerabilities.
- **Automation:** A *cron job* will be configured via *crontab -e* to trigger a system reboot every 2 years, maintaining long-term stability. This will be verified using *crontab -l*.
- **System Awareness:** A custom Message of the Day (MOTD) will be configured to display server information and usage warnings upon login.
- **Additional Packages:** A command-line tool not included by default, such as *mutt*, will be installed to demonstrate repository management and for command-line system notifications and reporting.

Demonstration Video

My video demo of this solution is about 15 minutes (total) long and follows this proposal's configuration plan. It uses troubleshooting tools like *journalctl* to demonstrate system health monitoring.

Links to Video Demos

1. <https://vimeo.com/1183543045/57a4107d2b?fl=tl&fe=ec>
2. <https://vimeo.com/1183551441/e3b38920ed?fl=tl&fe=ec>

APPENDIX

Table 1: Disk Partition Layout

Mount Point	Purpose	Filesystem	Initial Size	Rationale
/ (Root)	OS and system files	ext4	10 GB	Standard stability.
/home	User profiles	ext4	20 GB	Isolates user data from system files.
/srv/data	Shared departmental files	ext4	50 GB	Designed for future expansion.
swap	Virtual memory / System stability	swap	2 GB	Secondary overflow area for the RAM to prevent system crashes during peak processing loads, ensuring the server remains stable even if physical memory is fully exhausted.

Table 2: List of commands

SN	Category	Command	Purpose/Verification
1	System Identity	hostnamectl	Confirms the server name and OS details.
2	Service Status	systemctl status [service]	Proves the file server role is installed and active.
3	Storage & Disks	lsblk	Displays block devices to show partition structure.
4	Storage Usage	df -h	Verifies that partitions are mounted and shows available space.
5	Network Config	ip a	Displays the Static IPv4 address and interface status.
6	Network Details	cat /etc/netplan/*.yaml	Shows the persistent network configuration settings.
7	User Identity	id [username]	Verifies a specific user's UID and primary group.
8	Group Membership	groups [username]	Shows all supplemental groups assigned to a user.
9	File Permissions	ls -l	Shows ownership and standard read/write/execute bits.
10	Advanced Access	getfacl [path]	Verifies ACLs, specifically for Carrie Shumaker's read-only access.
11	Firewall Status	ufw status	Confirms the firewall is active and lists allowed ports.
12	Maintenance	apt update && apt upgrade	Demonstrates the process for keeping the system secure.
13	Automation	crontab -l	Displays the scheduled reboot job (every 2 years).
14	Package Verification	mutt --version	Confirms successful installation of a non-default package.
15	Troubleshooting	journalctl -xe	Demonstrates the ability to view system logs for stability.

Overall Benefits

1. Operational Reliability & Scalability

- **Preventive Storage Design:** Using a structured disk partition layout (separating `/`, `/home`, and `/srv/data`) guarantees that if one area fills up, the core operating system remains functional.
- **Future-Proofing:** The dedicated `/srv/data` partition is specifically designed for independent expansion as the company's data needs grow beyond the initial 38 employees.
- **Infrastructure Consistency:** Utilizing a Static IPv4 address ensures that all departments have a consistent, reliable connection point to the server, preventing the downtime associated with dynamic IP changes.

2. Enhanced Security & Access Control

- **Granular Permissions:** The use of Access Control Lists (ACLs) allows for complex permission structures, such as granting CIO Carrie Shumaker read-only access without compromising the security of departmental write-privileges.
- **Layered Defense:** Security is integrated through the Uncomplicated Firewall (UFW), regular automated package updates, and the use of non-root users for daily administration.
- **Auditability:** Organizing users into departmental groups (Operations, CRM, etc.) that mirror the organizational chart ensures clear accountability and easier management of system access.

3. System Stability & Automation

- **Automated Maintenance:** The 2-year scheduled reboot via cron maintains long-term kernel stability and clears system buffers without requiring manual intervention from IT staff.
- **Resource Management:** The allocation of 2 GB of Swap space acts as a vital safety net, preventing system crashes during peak processing loads by providing a secondary overflow for physical RAM.
- **System Awareness:** Custom MOTD configurations ensure that every user is greeted with essential server status and legal usage warnings immediately upon login.

4. Cost-Efficiency & Flexibility

- **Open-Source Advantage:** The choice of Ubuntu Linux ensures that PromisedLand avoids expensive proprietary licensing fees while gaining access to a massive ecosystem of professional-grade tools like *mutt* and *ufw*.
- **Cloud Agility:** Hosting the deployment on AWS EC2 allows for rapid deployment and the ability to scale hardware resources up or down based on real-time business requirements.

Call To Action

As your system administrator, I am ready to transition this Linux deployment from a strategic proposal to a high-performance reality. With this secure, scalable architecture, we will provide PromisedLand.com with the rock-solid infrastructure needed to empower our 38 employees and support our continued growth across North America. Let's build the future of our Independence data center today